

The irrationality exponent of $\log 2$

Detailed proof notes

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Abstract

We describe an argument for $\mu(\log 2) = 2$. Weighted interpolation produces a nonzero rational determinant. Arithmetic and analytic estimates for that same determinant contradict each other after the parameters are chosen. The statements below separate the geometric input, the two estimates, and the final deduction.

1. Main statement and proof dependencies

The conjecture on the irrationality exponent of $\log 2$ states the following.

Conjecture

The natural logarithm of 2 has irrationality exponent 2.

$$\mu(\log 2) = 2$$

These notes present a proof giving an affirmative answer. Throughout, $\lambda = \log 2$ and $\log$ is the natural logarithm. For an irrational real number x , $\mu(x)$ is the supremum of the real ν for which $0 < |x - p/q| < q^{-\nu}$ holds for infinitely many reduced fractions with $q \geq 2$.

Theorem 1.1 (Main statement). *For every real $\nu > 2$ there is an integer $Q \geq 2$ such that*

$$\left| \lambda - \frac{p}{q} \right| > q^{-\nu} \quad (p \in \mathbb{Z}, \ q \in \mathbb{Z}, \ q \geq Q). \quad (1.1)$$

Consequently, $\mu(\log 2) = 2$.

The fractions in (1.1) need not be reduced, and Q may depend on ν . The proof of Theorem 1.1 is completed in Section 9.

Where each result is proved

Sec.	Result established	Inputs used
2	Lemma 2.1: local multiplicity comparison	Transverse coordinates and weighted Bézout
3	Proposition 3.1: curve contact bound	Lemma 2.1 and separated weights
4	Theorem 4.1: weighted jet surjectivity	Proposition 3.1, blow-up positivity, vanishing
5	Propositions 5.1 and 5.2: nonzero minor and lower bound	Theorem 4.1 and denominator clearing
6	Lemmas 6.1 and 6.2: row expansion and growth	The matrix fixed in Proposition 5.1
7	Lemma 7.1 and Proposition 7.2: analytic upper bound	Lemmas 6.1 and 6.2; Taylor cancellation
8	Proposition 8.1: compatible parameter choices	Requirements of Theorem 4.1 and both estimates
9	Proof of Theorem 1.1	Propositions 5.2, 7.2 and 8.1; Dirichlet approximation

Sections 5-7 give estimates for fixed admissible data. Section 8 constructs such data under the contradiction hypothesis. Only then does H tend to infinity. Equation numbers identify formulas used elsewhere in the argument; other displayed calculations are left unnumbered.

2. Weighted spaces and a local comparison

This section fixes the interpolation spaces and proves Lemma 2.1, the local input to Proposition 3.1. Geometry is over $\mathbb{C}$. A curve or component is integral; multi-indices have nonnegative integer entries.

For positive rational weights $W = (w_0, \dots, w_m)$ and $V = (v_0, \dots, v_m)$, put

$$w\alpha = \sum_{i=1}^m w_i \alpha_i, \quad \mathcal{P}_W(H) = \text{span}_{\mathbb{C}} \{ Y^h X^\alpha : w_0 h + w\alpha \leq H \}.$$

$$\mathcal{J}_V(H) = \mathbb{C}[[t, u_1, \dots, u_m]] / (t^s u^\beta : v_0 s + \sum_i v_i \beta_i \geq H).$$

Thus the retained jet coefficients have weight strictly less than H . The fixed numerical assumptions for Sections 2-4 are

$$m, K \geq 1, \quad 0 < \theta < 1, \quad K \frac{w_0}{v_0} \theta^m < 1, \quad v_i = \frac{w_i}{\theta} \quad (1 \leq i \leq m). \quad (2.1)$$

The positive rational weights $w_1, \dots, w_m$ will be chosen successively in Section 3. The interpolation centers are $a_j = (y_j, c_{j1}, \dots, c_{jm})$, $0 \leq j < K$, with distinct nonzero y_j . The other coordinates are arbitrary.

Lemma 2.1 (Weighted local multiplicity). *Give affine coordinate directions positive rational degree weights ρ_a , and a commuting analytic frame positive costs κ_b . Let $N, \varepsilon > 0$. Let Z be a codimension- k component of the common zero set of polynomials of weighted degree at most N . Suppose every frame derivative of each defining polynomial of cost less than εN vanishes on Z . If A is a coordinate normal basis and B is a frame normal basis, both at a general smooth point of Z , then*

$$\prod_{a \in A} \rho_a \leq 2k! \varepsilon^{-k} \prod_{b \in B} \kappa_b. \quad (2.2)$$

Proof. Here a normal basis means a basis of the ambient tangent space modulo the tangent space of Z . Choose a smooth point where both bases are valid and which lies on no other component. Commuting flows in the B directions give transverse parameters b . The derivative hypothesis removes every coefficient whose B -weight is less than εN .

Restrict to the transverse affine slice obtained by fixing the coordinates outside A . The b parameters are still local coordinates on this slice. Replacing the tangential coordinates by analytic functions of b does not decrease the weight of a surviving monomial. The restricted ideal therefore lies in the ideal of monomials of B -weight at least εN . Its local quotient has length at least

$$\#\{\alpha \in \mathbb{Z}_{\geq 0}^k : \sum_{b \in B} \kappa_b \alpha_b < \varepsilon N\} \geq \frac{(\varepsilon N)^k}{2k! \prod_{b \in B} \kappa_b}.$$

For the counting inequality, assign each point of the open weighted simplex to the integer vector obtained by taking its coordinatewise floor. The corresponding unit cubes cover that simplex. In fact this gives the stronger bound without the factor 2.

The common zero on the slice is isolated. Choose k generic constant linear combinations of the restricted equations with an isolated common zero there. Their quotient length bounds the original quotient length from above. Weighted Bézout bounds this intersection length by $N^k / \prod_{a \in A} \rho_a$. One can obtain this bound from ordinary Bézout by clearing weight denominators and replacing each slice coordinate by a translated power; choose the translation so that the preimages of the point are unramified. Comparing the length bounds and cancelling N^k gives (2.2). This proves Lemma 2.1, uniformly in the equations and in Z . $\square$

3. The curve contact bound

We prove Proposition 3.1 from Lemma 2.1. Its conclusion is the positivity input to Theorem 4.1.

By (2.1), choose a positive rational σ sufficiently small that

$$(1 + 3\sigma)^{m+1} K \frac{w_0}{v_0} \theta^m < 1, \quad (1 + \sigma)\theta < 1. \quad (3.1)$$

Set $L_0 = m + 2$. Choose the weights successively so that, for subsets A, B of $\{0, \dots, m\}$ of the same size between 1 and m ,

$$\prod_{a \in A} w_a > C_* \prod_{b \in B} v_b, \quad C_* := \max_{1 \leq k \leq m} 2k! \left(\frac{L_0}{\sigma}\right)^k, \quad (3.2)$$

whenever the largest positive index in their symmetric difference belongs to A . In the ratio in (3.2), larger indices present in both sets contribute only $w_i/v_i = \theta$. The largest differing weight occurs only in the numerator. Thus a lower threshold for that weight depends only on earlier weights and the fixed data. There are finitely many requirements, all independent of the centers.

On the smooth complete model of a curve C meeting the affine chart, define

$$\deg_W C = \sum_P \max \left(0, \frac{-\text{ord}_P Y}{w_0}, \frac{-\text{ord}_P X_1}{w_1}, \dots, \frac{-\text{ord}_P X_m}{w_m} \right).$$

At a branch P over a_j , set $t = Y/y_j - 1$ and

$$h_P = \min \left(\frac{\text{ord}_P t}{v_0}, \min_{1 \leq i \leq m} \frac{\text{ord}_P (X_i - c_{ji} - \log(1+t))}{v_i} \right).$$

The logarithm is its formal expansion at that center. An identically zero function has order $+\infty$. The sum below includes every branch over every center.

Proposition 3.1 (Curve contact inequality). *For the weights chosen using (3.1) and (3.2), every such curve satisfies*

$$\deg_W C \geq (1 + \sigma) \sum_{P \mapsto a_j} h_P. \quad (3.3)$$

Proof. Step 1: excess contact gives vanishing. If (3.3) failed, the sum of contacts would be positive. For every sufficiently large N , the first inequality in (3.1) compares the leading counts of polynomial coefficients and jet conditions. It supplies a nonzero F_N of W -degree at most N with jet order at least $(1 + 3\sigma)N$ at all centers. Independence of those conditions is not needed for this dimension count.

$$D_0 = Y\partial_Y + \sum_{i=1}^m \partial_{X_i}, \quad D_i = \partial_{X_i} \quad (1 \leq i \leq m).$$

This commuting frame preserves the degree bound N . In the local jet coordinates, $D_0 = (1+t)\partial_t$ and $D_i = \partial_u$, so a derivative of cost at most σN retains jet order at least $(1 + 2\sigma)N$. On C its total zero order is at least $(1 + 2\sigma)N \sum h_P$, whereas its total pole degree is at most $N \deg_W C$. Failure of (3.3) makes the former larger. The zero-pole degree equality forces

$$D^\gamma F_N|_C = 0 \quad \left(\sum_{a=0}^m v_a \gamma_a \leq \sigma N \right). \quad (3.4)$$

Step 2: find a persistent component. Put $\delta_N = \sigma N/L_0$. In $Y \neq 0$, let V_r be the common zero set of derivatives of F_N of cost at most $r\delta_N$, for $0 \leq r \leq L_0$. By (3.4), all these sets contain the nonempty part C° of C in $Y \neq 0$. Start with a component of V_L containing C° , and extend backwards to nested components of the preceding sets. Their dimensions lie between 1 and m , since F_N is nonzero. Two adjacent components must coincide; call this component Z , of codimension k .

The equations from the earlier level have degree at most N . Commutativity implies that every additional derivative of cost less than δ_N vanishes on Z . Apply Lemma 2.1 with $\varepsilon = \sigma/L_0$. Equation (2.2) and the definition of C_* in (3.2) give

$$\prod_{a \in A} w_a \leq C_* \prod_{b \in B} v_b \quad (3.5)$$

for every coordinate normal basis A and frame normal basis B of Z .

Step 3: the comparison forces Y to be constant. Suppose some positive coordinate direction is not tangent to Z , and choose its largest index i . Some coordinate normal basis contains i . If a frame normal basis omitted i , its comparison with that coordinate basis would contradict (3.2) and (3.5): larger positive directions are tangent and occur in neither basis. Hence every frame normal basis contains D_i . The other frame directions span the ambient hyperplane $\ker(dX_i - dY/Y)$. Their failure to span the normal quotient says that the tangent space of Z lies in this hyperplane. Thus $dX_i = dY/Y$ on Z .

If Y were nonconstant on Z , take an algebraic curve in Z on which Y is nonconstant and pass to its smooth complete model. The differential dY/Y has a nonzero residue at a zero or pole of Y , while the exact differential dX_i has zero residues. This is impossible. If instead all positive coordinate directions are tangent to Z , they span its tangent space, because Z is proper; then $dY = 0$ directly. In either case Y is constant on Z and therefore on C .

Step 4: use the distinct Y fibers. Such a curve meets at most one center. Choose a nonconstant X_i on it. Since $\log(1+t) = 0$ on this constant fiber, zero-pole degree equality gives

$$\sum h_p \leq \frac{\deg(\text{poles } X_i)}{v_i} \leq \theta \deg_W C.$$

The second inequality in (3.1) now implies $\deg_W C \geq \theta^{-1} \sum h_p > (1 + \sigma) \sum h_p$, contradicting the assumed failure of (3.3). This completes Proposition 3.1. $\square$

4. From contact to interpolation

This section proves Theorem 4.1 using Proposition 3.1. It converts a bound on every curve into simultaneous prescription of all retained jet coefficients.

Theorem 4.1 (Weighted interpolation at distinct Y centers). *Assume (2.1), and choose the separated weights as in Section 3. For distinct nonzero y_j and arbitrary c_{ji} , the map*

$$P \longmapsto (P(y_j(1+t), c_{j1} + u_1 + \log(1+t), \dots, c_{jm} + u_m + \log(1+t)))_{j=0}^{K-1} \quad (4.1)$$

from the polynomial space to the direct sum of the K jet spaces is surjective for all sufficiently large integer multiples H of a common scale. The scale depends only on the weights. The lower threshold for H may depend on the centers; the thresholds for choosing the weights do not.

Proof. Step 1: algebraic realization. Choose a positive integer R_0 with all R_0/w_a and R_0/v_a integral. Embed affine space by the monomials of W -degree at most R_0 , including 1. Let X be its projective closure and L its hyperplane bundle. Pure powers of each coordinate occur among these monomials. Computing the maximum pole order of the embedding coordinates gives

$$\deg(L|_C) = R_0 \deg_W C. \quad (4.2)$$

At each center replace $\log(1+t)$ by a polynomial $G_i(t)$ whose remainder has weight greater than v_i . The triangular change $u'_i = X_i - c_{ji} - G_i(t)$ and its inverse preserve the filtration. They also preserve the minimum defining h_p : the remainder has order greater than v_i times that minimum. Set

$$I_j = (t^{R_0/v_0}, (u'_1)^{R_0/v_1}, \dots, (u'_m)^{R_0/v_m}). \quad (4.3)$$

These local ideals give a coherent ideal sheaf I supported at the centers. Equation (4.3) gives branch order $R_0 h_p$. On the blow-up $p: X' \rightarrow X$, write $I \mathcal{O}_{X'} = \mathcal{O}_{X'}(-E)$ and $A = p^* L$.

Step 2: positivity. Equations (3.3), (4.2) and (4.3) show that $A - (1 + \sigma)E$ has nonnegative degree on every curve whose image meets the affine chart. Boundary curves avoid E , and $-E$ has positive degree on contracted curves. Thus this divisor is nef. Relative ampleness of $-E$ gives $a_0 A - E$ ample for a sufficiently large integer $a_0 > 1$. Now

$$\begin{aligned} A - E &= \frac{a_0 - 1}{a_0(1 + \sigma) - 1} (A - (1 + \sigma)E) \\ &\quad + \frac{\sigma}{a_0(1 + \sigma) - 1} (a_0 A - E). \end{aligned} \quad (4.4)$$

Both coefficients in (4.4) are positive. The sum of an ample divisor and a nef divisor, with positive coefficients, is ample. Hence $A - E$ is ample.

Step 3: vanishing gives jets. Eventual recovery from the Rees algebra and relative Serre vanishing give $p_* \mathcal{O}_{X'}(-nE) = I^n$ and zero higher direct images for large n . Projection formula, Leray and ampleness from (4.4) then give

$$H^1(X, I^n \otimes L^n) = H^1(X', \mathcal{O}_{X'}(n(A - E))) = 0. \quad (4.5)$$

The exact sequence for I^n and (4.5) make global sections of L^n surject onto the local quotients $\mathcal{O}/I^n$. By (4.3), every monomial in I^n has weight at least nR_0 . Therefore each quotient maps onto the desired jet space with $H = nR_0$. For large n , sections of L^n come from degree- n homogeneous polynomials in the embedding coordinates. Their affine restrictions have W -degree at most H . Undoing the filtration-preserving change proves surjectivity of (4.1), and hence Theorem 4.1. $\square$

5. A nonzero minor and its arithmetic lower bound

Proposition 5.1 applies Theorem 4.1 to a rational matrix. Proposition 5.2 then bounds a fixed nonzero minor from below. Sections 6-7 will use exactly this minor.

Fix $\nu > 2$ and rational approximations $r_i = p_i/q_i$, $q_i \geq 2$, with $| \lambda - r_i | \leq q_i^{-\nu}$. Put $w_i = \lceil \log q_i \rceil$, $w_* = \min w_i$, and assume these weights satisfy Theorem 4.1. For rational $F > 2/\theta$, define

$$T_i = \left\lceil \frac{Fw_i}{v_0} \right\rceil, \quad G_i(t) = \sum_{1 \leq k < T_i} \frac{(-1)^{k+1}}{k} t^k. \quad (5.1)$$

Columns are indexed by (h, α) with $w_0 h + w\alpha \leq H$. Rows are indexed by $\rho = (j, s, \beta)$, with $0 \leq j < K$ and $v_0 s + w\beta/\theta < H$. The rational matrix entry is

$$[t^s u^\beta] (2^j (1+t))^h \prod_{i=1}^m (jr_i + G_i(t) + u_i)^{\alpha_i}. \quad (5.2)$$

Proposition 5.1 (Existence of the common determinant). *For every sufficiently large allowed H , the matrix (5.2) has a nonzero rational square minor Δ_H using every row. Write $M = M_H$ for its size.*

Proof. Equation (5.1) gives $v_0 T_i \geq Fw_i > v_i$. Thus the truncation change preserves the weighted jet filtration used in Theorem 4.1. Its centers here have distinct Y coordinates 2^j . Theorem 4.1 makes (5.2) surjective over $\mathbb{C}$. It consequently has a nonzero maximal minor; that minor is rational because all matrix entries are rational. This proves Proposition 5.1. $\square$

Fix one such minor for each H .

Define its average row weight and the denominator factor by

$$\bar{b} = \frac{1}{MH} \sum_{\rho=(j,s,\beta)} w\beta, \quad 0 \leq \bar{b} < \theta, \quad (5.3)$$

$$L_i = \text{lcm}(1, \dots, T_i - 1), \quad D_H = \prod_i L_i^{\lfloor H/w_i \rfloor}.$$

The row condition gives the last inequality in (5.3). An empty least common multiple is 1. Set $\Lambda = 4 \log 2$ and

$$E_{\text{ar}} = \frac{\Lambda F m}{v_0} + \Lambda \sum_i \frac{1}{w_i} + \frac{\theta}{w_*}. \quad (5.4)$$

Proposition 5.2 (Arithmetic lower bound). *Every nonzero minor fixed in Proposition 5.1 satisfies*

$$\frac{\log |\Delta_H|}{MH} \geq -(1 - \bar{b}) - E_{\text{ar}}. \quad (5.5)$$

Proof. Multiply column (h, α) by $\prod q_i^\alpha$, row (j, s, β) by $\prod q_i^{-\beta}$, and every entry by D_H . In (5.2), the remaining powers of each q_i are nonnegative, $q_i r_i = j p_i$ is integral, and L_i clears G_i . The resulting matrix is integral. Its determinant is nonzero by Proposition 5.1, so

$$\log |\Delta_H| \geq -M \log D_H - \sum_{\text{columns } i} \sum \alpha_i \log q_i + \sum_{\text{rows } i} \sum \beta_i \log q_i. \quad (5.6)$$

The column condition bounds the column cost in (5.6) by MH . The inequalities $w_i - 1 \leq \log q_i \leq w_i$, together with (5.3), bound its row saving below by $MH \bar{b} - MH\theta/w_*$. Finally, $\log \text{lcm}(1, \dots, n) \leq \Lambda n$ and (5.1) give $\log D_H/H \leq \Lambda F m/v_0 + \Lambda \sum 1/w_i$. Substitute these three bounds into (5.6). The definition (5.4) gives (5.5), proving Proposition 5.2. $\square$

6. The analytic row expansion

Keep the data and minor of Proposition 5.1 fixed. Lemma 6.1 rewrites its rows and bounds the scalar coefficients. Lemma 6.2 bounds the common entire functions; both feed into Section 7.

For a selected column $P = Y^h X^\alpha$, put

$$f_{a,P}(z) = [u^a] P(e^z, z + u_1, \dots, z + u_m). \quad (6.1)$$

This is $\text{binom}(\alpha, a) e^{hz} z^{|\alpha| - |a|}$ when $a \leq \alpha$, and zero otherwise. All row expansions may use the same finite set $S_H = \{a : wa \leq H\}$; outside it, every selected column function vanishes.

Write $\varepsilon_{ji} = j(r_i - \lambda)$, $\tau_i(t) = G_i(t) - \log(1 + t)$, and $z = j\lambda + \log(1 + t)$. Define the row vector $R_{j,\ell,a} = ([t^\ell] f_{a,P}(j\lambda + \log(1 + t)))_P$.

Lemma 6.1 (Exact expansion and scalar estimate). *Each row of (5.2) has the following expansion, with a restricted to S_H :*

$$\text{row}_{j,s,\beta} = \sum_{a \in S_H} \sum_{a \geq \beta} \sum_{0 \leq d \leq a - \beta} \sum_{k=0}^s \xi_{j,\beta,a,d,k} R_{j,s-k,a}, \quad (6.2)$$

$$\xi_{j,\beta,a,d,k} = \binom{a}{\beta} \binom{a-\beta}{d} \varepsilon_j^{a-\beta-d} [t^k] \prod_i \tau_i(t)^{d_i}. \quad (6.3)$$

The coefficients are independent of P . For every index in (6.2),

$$|\xi_{j,\beta,a,d,k}| \leq \exp \{ -\nu w(a - \beta) + H E_{\text{tr}} \}, \quad (6.4)$$

$$E_{\text{tr}} = \frac{\nu}{F} + \frac{\log 2}{v_0} + \frac{\log 4 + \log(2K) + \nu}{w_*}. \quad (6.5)$$

Proof. Since $e^{j\lambda} = 2^j$, substitution in (5.2) gives $Y = e^z$ and $X_i = z + u_i + \varepsilon_{ji} + \tau_i(t)$. Expand (6.1) first in $u + \varepsilon + \tau$ and then select u^β . The binomial expansion and coefficient convolution give exactly (6.2) and (6.3). This also proves column independence.

For a nonzero tail coefficient in (6.3), $\sum_i d_i T_i \leq k \leq s < H/v_0$. Equation (5.1) gives $wd < H/F$. On $|t| = 1/2$ the logarithmic tails have modulus at most 1, so Cauchy's estimate bounds that coefficient by 2^k . The approximation hypothesis gives $|\varepsilon_{ji}| \leq 2K e^\nu e^{-\nu w}$. Also $|a| \leq H/w_*$, and the product of the two binomial coefficients in (6.3) is at most $4^{|a|}$. Combining these bounds gives

$$\log |\xi| \leq -\nu w(a - \beta) + \nu wd + k \log 2 + |a|(\log 4 + \log(2K) + \nu).$$

Insert $w_d < H/F$, $k < H/v_0$, and $|a| \leq H/w_*$. The result is (6.4) with (6.5). Zero coefficients satisfy the bound directly. This proves Lemma 6.1. Since all indices in (6.2) range over sets of polynomial size in H , the number Q_H of terms per row can be bounded by a polynomial in H for fixed data. $\square$

Lemma 6.2 (Uniform growth of the column functions). *On $|z| = R_1 = 100K$, every function in (6.1) satisfies*

$$|f_{a,p}(z)| \leq B_H := \exp \left\{ H \left(\frac{100K}{w_0} + \frac{\log(200K)}{w_*} \right) \right\}. \quad (6.6)$$

Proof. Use $\text{binom}(\alpha, a) \leq 2^{|\alpha|}$, $h \leq H/w_0$, $|a| \leq H/w_*$, and $R_1 \geq 1$ in the explicit formula following (6.1). This gives $e^{hR} (2R_1)^{|\alpha|} \leq B_H$, which is (6.6). The bound is independent of a and of the selected columns. This proves Lemma 6.2. $\square$

7. Taylor cancellation and the analytic upper bound

Lemma 7.1 applies Lemma 6.2 to one determinant term from (6.2). Proposition 7.2 then incorporates the scalar estimate of Lemma 6.1 and sums all terms.

Lemma 7.1 (Uniform determinant bound from Taylor collisions). *Choose one row vector $R_{j,\ell,a}$ from each row expansion (6.2), and let T be the resulting M by M matrix. Let n_a count its rows with transverse index a . Then*

$$|\det T| \leq \exp \left\{ -c \sum_a n_a^2 + MH(E_{\text{hol}} + \rho_H) \right\}, \quad (7.1)$$

$$c = \frac{\log 2}{4}, \quad E_{\text{hol}} = \frac{100K}{w_0} + \frac{\log 2}{v_0} + \frac{\log(200K)}{w_*}. \quad (7.2)$$

Here ρ_H tends to zero for fixed data, uniformly over all row choices and selected columns.

Proof. Expand $f_{a,p}(z) = \sum_{d \geq 0} b_{a,d,p}(z/R_1)^d$. Lemma 6.2 and Cauchy's estimate give $|b_{a,d,p}| \leq B_H$. On $|t| = 1/2$, $|j\lambda + \log(1+t)| < R_1/2$. Therefore the row functional applied to $(z/R_1)^d$ has modulus at most $2^\ell 2^{-d}$.

If two rows choose the same pair (a, d) , their coefficient vectors $(b_{a,d,p})_p$ coincide and that determinant vanishes. Every surviving assignment thus satisfies

$$\sum_{\text{rows}} d \geq \sum_a \binom{n_a}{2} = \frac{1}{2} \left(\sum_a n_a^2 - M \right). \quad (7.3)$$

For each assignment, the coefficient determinant has modulus at most $M! B_H^M$. Split $2^{-\sum d}$ into two equal factors. Bound the first with (7.3), and sum the second over all nonnegative degrees, giving C_0^M , where $C_0 = (1 - 2^{-1/2})^{-1}$. Since $\sum \ell \leq MH/v_0$, absolute convergence justifies the expansion and gives

$$|\det T| \leq M! B_H^M 2^{MH/v_0} C_0^M \exp \left\{ -\frac{\log 2}{4} \left(\sum_a n_a^2 - M \right) \right\}.$$

Take logarithms and use (6.6). This is (7.1) with (7.2) and, for example,

$$\rho_H = \frac{\log(M!) + M \log C_0 + cM}{MH} \leq \frac{\log M + \log C_0 + c}{H}. \quad (7.4)$$

M grows polynomially in H , so (7.4) tends to zero. It depends on M and H , not on the row choices or columns. This proves Lemma 7.1. $\square$

For a cutoff $a \in (0, 1)$ and proportion $\eta \in (0, 1)$, use α' for a transverse multi-index and define

$$D_a(H) = \#\{\alpha' \in \mathbb{Z}_{\geq 0}^m : w\alpha' \leq aH\}, \quad L_H = \frac{\eta^2 M}{HD_a(H)}.$$

Counting lattice points in the fixed weighted simplices gives

$$M \sim \frac{K\theta^m H^{m+1}}{(m+1)! v_0 \prod_i w_i}, \quad D_a(H) \sim \frac{(aH)^m}{m! \prod_i w_i}. \quad (7.5)$$

Dividing the two asymptotics in (7.5) yields

$$L_H \longrightarrow L_* := \frac{\eta^2 K \theta^m}{(m+1) v_0 a^m}. \quad (7.6)$$

Proposition 7.2 (Analytic upper bound for the same minor). *For Δ_H fixed in Proposition 5.1, let $E_{an} = E_{tr} + E_{hol}$ using (6.5) and (7.2). Then*

$$\frac{\log|\Delta_H|}{MH} \leq E_{an} + o(1) + \max\{-cL_H, -\nu(a(1-\eta) - \bar{b})\}. \quad (7.7)$$

The error tends to zero after all weights, centers and scalar parameters have been fixed, uniformly in the choice of the nonzero minor.

Proof. Expand the determinant using (6.2). By the column independence in Lemma 6.1, each term is a product of scalars (6.3) times a determinant T covered by Lemma 7.1.

If at least ηM chosen rows have $w\alpha' \leq aH$, there are at most $D_a(H)$ such transverse indices. Cauchy-Schwarz gives $\sum_{\alpha'} n_{\alpha'}^2 \geq (\eta M)^2 / D_a(H) = MHL_H$. In (6.4), every $w(\alpha' - \beta)$ is nonnegative. Drop that saving and apply (7.1); the normalized logarithm of the term is at most $E_{an} + \rho_H - cL_H$.

Otherwise more than $(1 - \eta)M$ rows have $w\alpha' > aH$. Equation (5.3) gives $\sum w(\alpha' - \beta) > MH(a(1 - \eta) - \bar{b})$. Use this saving in (6.4), and drop the nonpositive collision term in (7.1). The normalized logarithm is at most $E_{an} + \rho_H - \nu(a(1 - \eta) - \bar{b})$.

There are at most Q_H^M terms, with Q_H polynomial in H by Lemma 6.1. Summing adds at most $\log Q_H/H$ to the normalized logarithm. Together with (7.4) this is $o(1)$. Taking the worse of the two bounds proves (7.7), and hence Proposition 7.2. $\square$

8. Choosing compatible parameters

Proposition 8.1 constructs data satisfying Theorem 4.1 while leaving a strict gap between the bounds in Propositions 5.2 and 7.2. This section makes the order of choices explicit.

Proposition 8.1 (Simultaneous interpolation and error margins). *Fix $\nu > 2$. Assume rational approximations with $|\lambda - p/q| < q^{-\nu}$ exist for unbounded q . Then one can fix m, K , all weights and centers, and a, η, F as above, satisfying Theorem 4.1, such that*

$$E_{ar} + E_{an} < \varepsilon_0 < g, \quad 1 + E_{ar} + E_{an} < 2 < cL_*, \quad (8.1)$$

where $g = \nu(a(1 - \eta) - \theta) - (1 - \theta) > 0$. These data remain fixed as H tends to infinity through the allowed multiples.

Proof. Step 1: scalar choices. Choose rational b_0 and then a sufficiently small positive rational δ so that

$$\frac{1}{2} < b_0 < 1 - \frac{1}{\nu}, \quad \theta = 1 - \delta, \quad a = 1 - b_0\delta.$$

The first strict inequality gives $a^2 < \theta$ for small δ ; the second gives $\nu(a - \theta) > 1 - \theta$. Also $0 < \theta < a < 1$. Consequently the following intervals are nonempty; choose rational C and B in them:

$$\frac{a}{\theta} < C < \frac{1}{a}, \quad a < B < \min\left(\frac{1}{C}, C\theta\right). \quad (8.2)$$

Choose $\eta > 0$ sufficiently small that $g > 0$. Set $\varepsilon_0 = \min(g/2, 1/2)$ and choose rational $F > 2/\theta$ with $\nu/F < \varepsilon_0/3$.

Step 2: choose the dimension. For increasing positive integers m , put

$$K = \lfloor C^m \rfloor, \quad w_0 = B^{-m}, \quad v_0 = 2K\theta^m w_0. \quad (8.3)$$

Equation (8.3) makes the interpolation ratio in (2.1) exactly $1/2$. Equation (8.2) gives $CB < 1$, $C\theta/B > 1$, and $B/a > 1$. Substituting (8.3) into (7.6) therefore gives

$$\frac{K}{w_0} \longrightarrow 0, \quad \frac{m}{v_0} \longrightarrow 0, \quad L_* = \frac{\eta^2(B/a)^m}{2(m+1)} \longrightarrow \infty.$$

Fix one sufficiently large finite m so that

$$\frac{\Lambda F m + 2\log 2}{v_0} + \frac{100K}{w_0} < \frac{\varepsilon_0}{3}, \quad cL_* > 2. \quad (8.4)$$

Now fix σ and all successive separation thresholds of Section 3. They depend on the data already chosen, not on the approximation centers.

Step 3: choose the approximations. The assumed unbounded set of good denominators permits choosing $q_1, \dots, q_m$ successively so that $w_i = \lfloor \log q_i \rfloor$ meets (3.2) and any prescribed common lower bound. Require in addition

$$\Lambda \sum_i \frac{1}{w_i} + \frac{\theta + \log 4 + \log(2K) + \nu + \log(200K)}{w_*} < \frac{\varepsilon_0}{3}. \quad (8.5)$$

Set $r_i = p_i/q_i$ and use the centers $(2^j, jr_1, \dots, jr_m)$. Theorem 4.1 and Proposition 5.1 now apply. No threshold for an earlier choice depends on these later centers.

Step 4: check the full error budget. Add (5.4), (6.5) and (7.2):

$$\begin{aligned} E_{\text{ar}} + E_{\text{an}} &= \frac{\nu}{F} + \frac{\Lambda F m + 2\log 2}{v_0} + \frac{100K}{w_0} + \Lambda \sum_i \frac{1}{w_i} \\ &\quad + \frac{\theta + \log 4 + \log(2K) + \nu + \log(200K)}{w_*}. \end{aligned} \quad (8.6)$$

In (8.6), ν/F is less than $\varepsilon_0/3$ by step 1, the next two terms together are less than $\varepsilon_0/3$ by (8.4), and the remaining terms are less than $\varepsilon_0/3$ by (8.5). Hence $E_{\text{ar}} + E_{\text{an}} < \varepsilon_0$. Since $\varepsilon_0 < g$ and $\varepsilon_0 \leq 1/2$, this and (8.4) imply both inequalities in (8.1). This proves Proposition 8.1. $\square$

9. Completion of the main theorem

Proof of Theorem 1.1. We use Propositions 5.2, 7.2 and 8.1. Fix $\nu > 2$ and suppose (1.1) fails. For every Q there is a denominator $q \geq Q$ with error at most $q^{-\nu}$. Proposition 8.1 therefore supplies fixed admissible data. For arbitrarily large allowed H , Proposition 5.1 supplies $\Delta_H \neq 0$.

Compare the two branches of the upper bound

Proposition 5.2 gives the lower bound (5.5), and Proposition 7.2 gives the upper bound (7.7), for that same Δ_H . In the first branch, (7.6) and the second inequality of (8.1) imply, for all sufficiently large H ,

$$E_{\text{an}} + o(1) - cL_H < -(1 - \bar{b}) - E_{\text{ar}}.$$

Indeed, $cL_* > 1 + E_{\text{ar}} + E_{\text{an}}$, and $1 - \bar{b} \leq 1$ by (5.3). The strict gap absorbs both $L_H - L_*$ and $o(1)$.

For the other branch, use $\bar{b} < \theta$ from (5.3) and $\nu > 2$ to obtain

$$\nu(a(1 - \eta) - \bar{b}) - (1 - \bar{b}) \geq \nu(a(1 - \eta) - \theta) - (1 - \theta) = g. \quad (9.1)$$

The first inequality of (8.1) leaves $g - E_{\text{ar}} - E_{\text{an}} > 0$. Equation (9.1) therefore makes the second branch of (7.7), including its $o(1)$, strictly smaller than (5.5) for large H as well. Both branches contradict the lower bound on the same nonzero determinant. The assumed unbounded family cannot exist. This proves (1.1).

Deduce the value of the irrationality exponent

If λ were rational, representing it with arbitrarily large denominators would contradict (1.1). Thus λ is irrational. Since (1.1) holds for every $\nu > 2$, it gives $\mu(\lambda) \leq 2$. Dirichlet approximation gives infinitely many reduced rational approximations with exponent 2, and hence $\mu(\lambda) \geq 2$. Therefore

$$\mu(\log 2) = 2.$$

This completes Theorem 1.1. □

Its conclusion allows Q to depend on ν ; it does not assert a uniform positive lower bound of the stronger form c/q^2 .

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AI usage disclosure

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